

## Lecture 2 - Density-independent deterministic growth

### Announcements:

Today - Paper discussion until  $\approx 10:30$ , then lecture.

Next class: Bring laptop with R, having read readings and problem set.

Will work on problem set together (intro. to R).

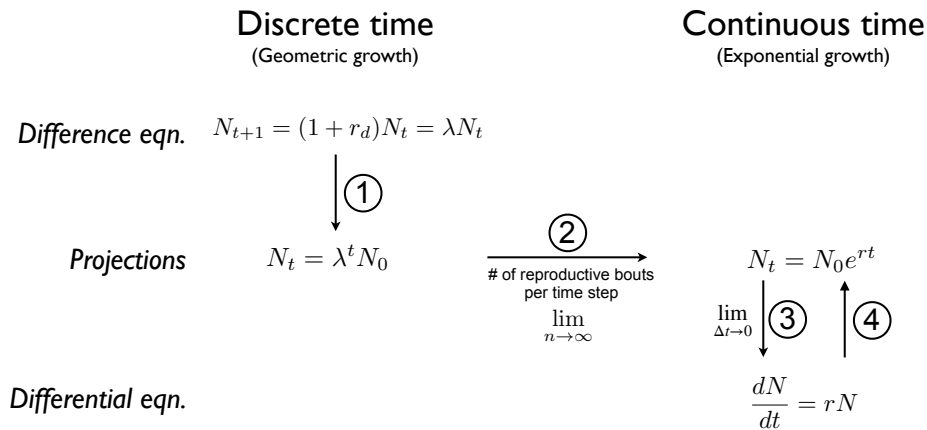
### Today's concepts:

Geometric vs. Exponential growth

Discrete vs. Continuous (Difference vs. Differential equations)

Population vs. Per capita rates of change

Overview [write out before lecture] :



① Simplest possible model: Discrete time difference equation:

$$N_{t+1} = N_t + B - D + I - E$$

$B$ -Total births;  $D$ -Total deaths;  $I$ -Immigration;  $E$ -Emigration

Let:  $I = E$ ,  $B = b_d N$ ,  $D = d_d N$ .

$b_d$  - births per individual

$d_d$  - deaths per individual per time step (i.e. probability of each individual dying)

Thus:

$$N_{t+1} = N_t + (b_d - d_d)N_t = (1 + b_d - d_d)N_t = (1 + r_d)N_t = \lambda N_t$$

$r_d$  - discrete growth factor/increment (Ted Case writes this as  $R$ ).

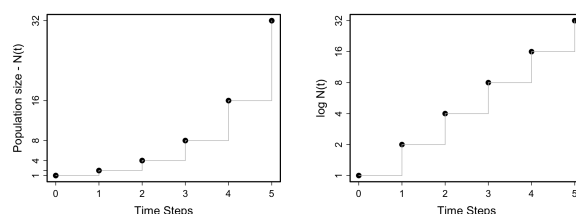
$\lambda$  - "finite rate of increase" - per capita rate of growth if population is growing geometrically.

Draw  $N(t)$  vs.  $t$  on arithmetic scale on board in steps (4 y-axis increments)

$$N(0) = 1, \lambda = 2 \implies N(t) = 1, 2, 4, 8, 16, 32, \dots \implies \text{Geometric growth}$$

NOTE: Not just doubling!  $\lambda$  can be any number!

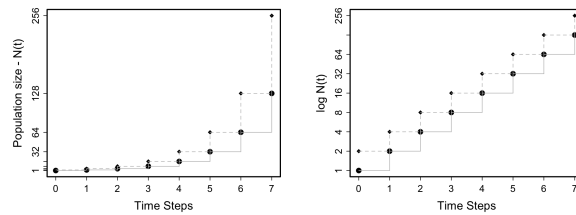
Then draw on log-scale. (6 y-axis increments)



Show R plots

Q: What happens if we start at different population size at same  $\lambda$ ?

Add points for  $N(0)=2$  on drawing. Plot in R



Q: Why linear on log-scale?

A: Logarithms convert multiplicative processes to additive processes.

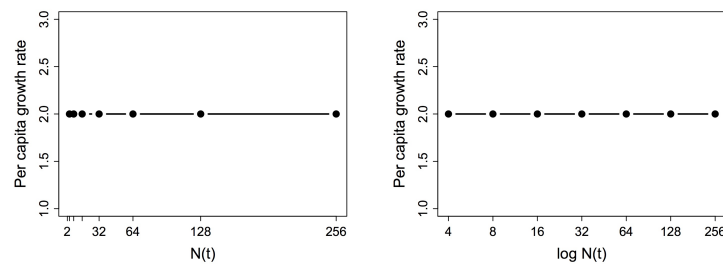
On log-scale, products become sums, ratios become differences:

$$\left. \begin{aligned} y &= a \cdot b \\ \log(y) &= \log(a \cdot b) = \log(a) + \log(b) \end{aligned} \right| \log(y) = \log\left(\frac{a}{b}\right) = \log(a) - \log(b) \quad \left| \quad y = \frac{a}{b} \right.$$

Q: Why do we call this density-independent population growth?

A: Density independence of *per capita* growth rate

Show plot of  $\lambda = \frac{N_{t+1}}{N_t}$  vs.  $N_t$



Want to predict  $N(T)$ : Analytical solution of  $\lambda \dots (\lambda(\lambda N_0)) = \lambda^T N_0$

Q: What have we assumed?

A: List includes:

- synchronous discrete reproduction and mortality
- constant growth rate (deterministic, i.e., non-stochastic)
- no density-dependence

Note:

Will go back and forth between calculating per capita growth rate as either  $\frac{N_{t+1}}{N_t}$  or  $\frac{(N_{t+1}-N_t)}{N_t}$ .

Why? Because:

$$N_{t+1} = \lambda N_t \implies \lambda = \frac{N_{t+1}}{N_t}$$

but also

$$N_{t+1} = \lambda N_t = (1 + r_d) N_t = N_t + r_d N_t \implies N_{t+1} - N_t = r_d N_t \implies \frac{N_{t+1} - N_t}{N_t} = r_d = \lambda - 1$$

## ② Discrete vs. continuous growth

Recovering the continuous from the discrete:

$$r_d = 0.5 \implies \lambda = 1.5$$

$$t = 1 \text{ year}$$

$$N_1 = \lambda N_0 = (1 + 0.5)N_0$$

$$t = \frac{1}{2} \text{ year}$$

$$N_1 = \lambda^2 N_0 = \left(1 + \frac{r_d}{2}\right)^2 N_0 = (1 + 0.25)^2 N_0$$

$$N_1 = \left(1 + \frac{r_d}{n}\right)^n N_0$$

$$\frac{N_1}{N_0} = \left(1 + \frac{r_d}{n}\right)^n$$

$$\lambda = \lim_{n \rightarrow \infty} \left(1 + \frac{r_d}{n}\right)^n = e^r \quad r - \text{instantaneous per capita growth rate}$$

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### R-demonstration: Euler's constant

Let:  $n = 1$ ,  $N_0 = 1$ ,  $r_d = 1$

True  $e = \exp(1) = 2.71828\dots$

Estimate  $e$  as  $\frac{N_1}{N_0}$  over increasing values of  $n$

$$\text{e.g., } n = 1 \Rightarrow \frac{N_1}{1} = \left(1 + \frac{r_d}{n}\right)^n$$

$$n = 1 \Rightarrow \frac{N_1}{1} = \left(1 + \frac{1}{1}\right)^1 = 2$$

$$n = 2 \Rightarrow \frac{N_1}{1} = \left(1 + \frac{1}{2}\right)^2 = 2.25$$

$$n = 3 \Rightarrow \frac{N_1}{1} = \left(1 + \frac{1}{3}\right)^3 = 2.30707\dots$$

$$n \rightarrow \infty \Rightarrow \frac{N_1}{1} = \left(1 + \frac{1}{n}\right)^n = 2.71828\dots$$

Per capita growth in continuous time is not  $r_d$  but rather  $e^r$  (with  $r < r_d$ ).

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$$\lim_{n \rightarrow \infty} \left(1 + \frac{r_d}{n}\right)^n = e^r$$

Definition of natural logarithm as the anti-exponential and  $e$  as the anti-log:  $\ln(e^x) = x$ .

Note that  $\log = \log_e = \ln$ .

The same definition holds for logarithms in other bases:

$$\log_{10}(10^x) = x \quad \log_2(2^x) = x$$

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Summarize  $r$  vs.  $r_d$  vs.  $\lambda$

$$\left(1 + \frac{r_d}{1}\right) = \lambda = e^r$$

And since  $\ln$  is the anti-exponential (i.e.  $\ln(e^x) = x$ ), we equivalently have

$$\ln(1 + r_d) = \ln(\lambda) = r$$

Thus another way to write population growth is...

$$N(T) = \lambda^T N_0 = N_0 e^{rT}$$

...which is now **exponential growth** / *continuous reproduction*

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Discussion of discrete vs. continuous as a spectrum depending on time-scale



Why emphasize this? Empirical measurements of real populations are intrinsically discrete (we measure  $N_0, N_1, N_2, \dots$ ). Many empiricists therefore (inappropriately) default to discrete time models to estimate parameters like  $\lambda$ , even when biology of species exhibits continuous growth on time-scales being considered (for which estimating parameters like  $r$  is appropriate for subsequent inferences).

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③ How to get instantaneous population-level growth rate from projection equation,  $N_0 e^{rt}$ ?  
That is, how do we show that:

$$\lim_{\Delta t \rightarrow 0} \left( \frac{\Delta N_t}{\Delta t} \right) = \frac{dN_t}{dt}$$

Need to take the derivative of  $N_0 e^{rt}$  with respect to time  $t$ . Use Product Rule:

$$\frac{d(XY)}{dt} = \frac{d(X)}{dt} \cdot Y + X \cdot \frac{d(Y)}{dt}$$

*“The derivative of a product is the sum of the product of the derivative of each term multiplied by the other term.”*

Thus:

$$\frac{d(N_0 \cdot e^{rt})}{dt} = \frac{d(N_0)}{dt} \cdot (e^r)^t + N_0 \cdot \frac{d((e^r)^t)}{dt}$$

Recall that derivative of a constant = 0 & derivative of  $a^x = \ln(a) \cdot a^x$ .

Thus:

$$\begin{aligned} \frac{d(N_0 e^{rt})}{dt} &= 0 \cdot (e^r)^t + N_0 \cdot \ln(e^r) \cdot (e^r)^t \\ &= N_0 \cdot r \cdot (e^r)^t \\ &= N_0 \cdot r \cdot e^{rt} \\ &= r \cdot N_0 e^{rt} \end{aligned}$$

Since  $N_t = N_0 e^{rt}$  for any time  $t$ ...

$$= r N_t = \frac{dN_t}{dt} \quad \dots \text{then drop } t \text{ subscript}$$

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④ Could also go in opposite direction from  $\frac{dN}{dt} \rightarrow N_0 e^{rt}$ :

$$\begin{aligned} \frac{dN}{dt} &= rN \\ \frac{1}{N} \frac{dN}{dt} &= r \\ \int_0^T \frac{1}{N} \frac{dN}{dt} dt &= \int_0^T r dt \quad (\text{Think of } T \text{ as a constant, and } t \text{ in } dt \text{ as a variable}) \\ \int_0^T \frac{1}{N} \frac{dN}{dt} dt &= rt \Big|_0^T = r \cdot T - r \cdot 0 = rT \end{aligned}$$

Using  $\int \frac{1}{x} dx = \ln(x)$ . That is,  $\int_{x_0}^{x_T} \frac{1}{x} dx = \ln(x_T) - \ln(x_0)$ , we have that

$$\ln(N_T) - \ln(N_0) = rT$$

$$\ln\left(\frac{N_T}{N_0}\right) = rT$$

$$\frac{N_T}{N_0} = e^{rT}$$

$$N_T = N_0 e^{rT}$$


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Q for class: Qualitative analysis of population *growth vs. decline*

| <i>Population change</i> | $\lambda$ | $r_d$ | $r$   |
|--------------------------|-----------|-------|-------|
| No change                | 1         | 0     | 0     |
| Growth                   | $> 1$     | $> 0$ | $> 0$ |
| Decline                  | $< 1$     | $< 0$ | $< 0$ |

Q: What are the units of  $r$ ? A: 'Individuals *per individual* per time'

Q: What are the units of population growth rate,  $\frac{dN}{dt}$ ? A: 'Individuals per time'

Q: What are the units of per capita growth rate,  $\frac{1}{N} \frac{dN}{dt}$ ? A: 'Individuals *per individual* per time'

### Polynomial representation of $\frac{dN}{dt}$

Recall from class 1:

Population size as function of time expressed using polynomial:

$$\begin{aligned}
 N(t) &= \sum_{n=0}^{\infty} \beta_n t^n \\
 &= \beta_0 + \beta_1 t + \beta_2 t^2 + \dots
 \end{aligned}$$

Thus, could also think of exponential growth as

$$\frac{dN}{dt} = f(N) = \sum_{n=0}^{\infty} \beta_n N^n$$

with

$$\beta_0 = 0, \quad \beta_1 = r, \quad \beta_{n>1} = 0.$$

Exponential growth is nothing more than a 'first-order' approximation!